

Decision log

Approach and priorities

I first cloned the repository and worked through the contracts, invoice fields and submission requirements. I started with H1 because its labels let me measure results. I considered embedding the contracts and retrieving evidence for invoice reasoning, but chose reusable JSON and deterministic checks as the foundation: extract rules once, make calculations traceable, and limit repeated AI reasoning.

With Codex assistance, I built the first H1 TXT-to-JSON parser and invoice checker. It answered only 40 of 908 eligible invoices, so I examined unanswered samples with Codex against the original clauses. I expanded both the JSON and checking code with supported rules for discounts, limits and related services, keeping terminology assumptions separate from contract facts. I also tried a small Voyage/Gemini retrieval pilot; later improvements reused some of its saved matches.

Next, I grouped remaining blockers into unclear descriptions, ambiguous units, patient history and volume discounts. I introduced targeted fallback: AI proposes description or clause interpretations, while code validates them and calculates the result. History and volume use possible usage ranges. I asked for general fixes to problem types, not exceptions for individual invoices, preserved earlier results, and compared again with H1 labels. Unsupported cases stayed unanswered.

After improving H1, I used a general AI extractor for each hospital's own source bundle, including amendments, and applied the shared workflow across hospitals. I reused cached responses and worked through blockers one type at a time. While improving one hospital, I found recurring patterns in others, such as compound units, abbreviated service families and uncertain historical quantities. I then re-read each hospital's own clauses before reusing a general solution; rules and rates were never transferred between hospitals. After the initial time-boxed approach, I requested further H2-H5 experiments, outside the stated 6-8-hour cap.

Decisions and unresolved alternatives

1. **Service identity can be incomplete.** H2 "ELECT CARD SVC" could mean Nursing Observation or Transport. Later outside-group references supported Transport, conditionally: at least three reference invoices, $\geq 80\%$ support and $\leq 20\%$ competing support, excluding the target's own price. This is not confirmed identity.
2. **A rate does not settle an ambiguous unit.** Four telemetry services say "per hour, per item." Later H2-H5 pilots used supplied numeric quantities under each contract's multiplication clause, with explicit assumptions and confidence capped at 0.50. Pricing opinions do not verify hours, items or physical quantities.
3. **Dates need explicit assumptions.** H2's day runs 07:00-06:59, but invoices lack timestamps. I treated service dates as billing-day labels, conditionally. Malformed dates are quarantined from dated patient-history checks, while their possible volume effects remain. This is not permission to ignore all date risks.
4. **An exclusion boundary has two readings.** "Within 10 days" could include or exclude day 10. I left exact-boundary exclusions unresolved, instead of treating the labels' inclusive reading as an explicit contract definition. The rule applies to the same patient; authoritative clarification remains needed.
5. **Uncertain history must not become confirmed usage.** I tracked minimum and maximum possible prior usage. Later family-based bounds assume family wording is truthful; H3/H5 reference-supported experiments retain numeric historical quantities despite wrong unit labels, conditionally, preserving those errors. Ordering uses service date then line ID; money uses contract-ordered half-up cents.
6. **Amendments and allocation need consistent policies.** I applied H3 repricing by service date and retained uncertain retrospective settlements. For H5, missing line facilities inherit the header facility. Repeated charges across distinct invoices are assigned to the earliest dated bill as a stated policy, not an explicit allocation clause. All 62 duplicate-ID records stay internal.
7. **An error need not reveal its correction.** A cap proves excess, not the true quantity. I flag proven errors without guessing totals. Unresolved verdicts stay outside the six-column submission; only 0/1 opinions are submitted. Confidence is an uncalibrated evidence score, not a measured probability; missing totals cap it at 0.50.
8. **H4's missing service mapping remains unresolved.** VST foc NEURO /CW-8682 occurs once in the supplied invoices; other contracts provide no reliable equivalent. Its 23075-cent price matches Comprehensive Urologic Imaging Interpretation, but that conflicts with the neurological-visit wording. Online searches also yielded no authoritative mapping. I did not transfer another hospital's rates or guess from price. All 69 remaining H4 volume-only cases depend on this line (four also have other uncertain lines). Request H4's code-to-service crosswalk or original service record; retain uncertainty meanwhile.

Review and disclosure

Codex assisted code, source review and writing; these were not independent expert reviews. The early pilot used Voyage retrieval/reranking and Gemini via OpenRouter; the current extractor and AI fallback use Gemini. H1 labels informed development, not independent validation. Next I would seek unit/boundary clarification and independently reviewed examples.

Sources: original contracts; H4_REVIEW.md; EVALUATION.md; prompts/README.md; AI_USAGE.md.